

**ILLINOIS INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE**

**Database Organization
Course CS-425 02**

Homework No.1

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Homework N°1

Class demonstration: Drawing ERD given business rules

- Using Chen's notation, draw an ERD according to the business rules below.
Translate the ERD to a logical level showing the relation schemas – include primary and foreign key fields.

Problem 1

Sheep farm feed supplier business rules:

1. All entities must have surrogate primary keys.
2. A contact person represents one or more suppliers, and each supplier has exactly one contact person.
1. A supplier supplies many feed products, and every feed product can be supplied by many suppliers.
2. Each feed product belongs to a specific feed type, and many feed products can have the same feed type.

Definition of entities

- Contact Person
 - Primary Key: CPID
 - Attributes: CName, CPhone, CEmail
- Supplier
 - Primary Key: SID
 - Attributes: SName, SAddress
 - Foreign Key: CPID
- Feed Product
 - Primary Key: FPID
 - Attributes: FPName, FPDescription
 - Foreign Key: FTID
- Feed Type
 - Primary Key: FTID

- Attributes: FTName, FTDescription
- Supply_feed_product
 - Primary Key: SFPID
 - Foreign keys: FPID, SID
 - Attributes: SFPQuantity

Relationships:

Represents

Between ContactPerson and Supplier 1:M

(One ContactPerson can represent many Suppliers, and each Supplier has exactly one ContactPerson)

Supply_feed_product (Bridge entity relationship)

Between Supplier and FeedProduct M

(Many Suppliers can supply many FeedProducts, and each FeedProduct can be supplied by many Suppliers)

Belongs

Between FeedProduct and FeedType M:1

(Many FeedProducts can belong to one FeedType, and each FeedProduct belongs to one FeedType)

Relation Schemas:

ContactPerson(CPID, CName, CPhone, CEmail)

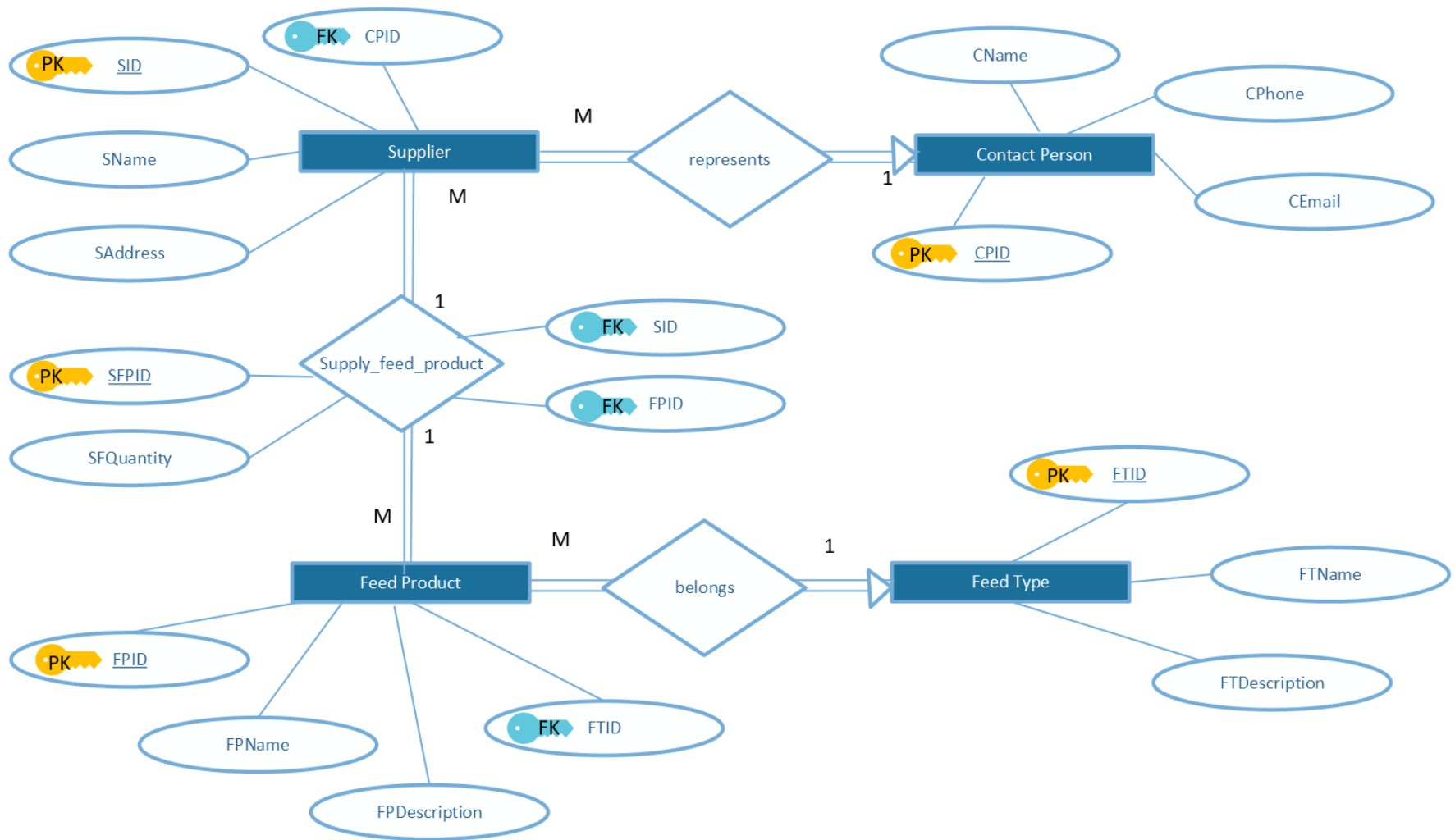
Supplier (SID, CPID, SName, SAddress)

Feed_Product(FPID, FTID, FPName, FPDescription)

FeedType(FTID, FTName, FTDescription)

Supply_feed_product(SFPID, FPID, SID, SFPQuantity)

ERD diagram



Problem 2

Launch Pad business rules:

1. A launchpad hosts multiple launches over time, and each launch is hosted by exactly one launchpad. Therefore, Launchpads will be created in the database even before the first launch is hosted there.
2. A launchpad is located in a country, and a country, in turn, is located on a continent.
3. Each launchpad has a name and a location.
4. A launch requires a launchpad that hosts it, a launch vehicle (the rocket), and the payload.
5. For a launch, the date, time and name should be recorded as well as whether it is a crewed launch or not.
6. Each launch is performed by a launch vehicle. The serial number for the launch vehicle should be recorded. A launch vehicle can perform multiple launches over time.
7. Each launch vehicle is of a specific launch vehicle type. The name, maximum thrust and whether it is reusable need to be recorded for the launch vehicle type.
8. A specific manufacturer makes each launch vehicle type, and a manufacturer can make many different launch vehicle types.
9. Each launch carries one payload, and a payload is only carried by one launch.
10. A manufacturer also makes each payload, and a manufacturer can make many different payloads over time.
11. Multiple crew members can be carried on a crewed launch, and each crew member can be carried on multiple launches over time.
12. Each crew member has a nationality (the country they come from), and their name and surname also need to be recorded.
13. Crew members can appear in the database before they launch for the first time.

- *NB: Draw an ERD representing these business rules.*
- *You are required to show relation schemas that include surrogate primary key and foreign key fields.*

Definition of entities

1. Launchpad:

Primary key: LaunchpadID

Attributes: Name, Location

Foreign key: CountryID

2. Country

Primary key: CountryID

Attributes: Name

Foreign key: ContinentID

3. Continent

Primary key: ContinentID

Attributes: Name

4. Launch

Primary key: LaunchID

Foreign keys: LaunchpadID , VehicleID , PayloadID

Attributes: Date, Time, Name, Crewed (Boolean)

5. Launch Vehicle

Primary key: VehicleID

Attributes: SerialNumber

Foreign key: VehicleTypeID

6. Launch Vehicle Type

Primary key: VehicleTypeID

Attributes: Name, MaxThrust, Reusable (Boolean)

Foreign Key: ManufacturerID

7. Manufacturer

Primary key: ManufacturerID

Attributes: Name

8. Payload

Primary key: PayloadID

Foreign Key: ManufacturerID

Attributes: Description

9. Crew Member

Primary key: CrewMemberID

Attributes: Name, Surname

Foreign Key: CountryID

10. Launch Crew (Junction Table for Many-to-Many relationship between Launch and Crew Member)

Foreign keys: LaunchID , CrewMemberID

Composite key (LaunchID, CrewMemberID) as PK

Relationships

- Launchpad is located in a Country, which in turn is located on a Continent.
- Each Launchpad hosts multiple Launches.
- A Launch requires a Launchpad, a Launch Vehicle, and a Payload.
- Launch Vehicle performs multiple Launches and is of a specific Launch Vehicle Type.
- Launch Vehicle Type is made by a Manufacturer.
- Each Payload is made by a Manufacturer and carried by one Launch.
- A Crewed Launch can carry multiple Crew Members, and each Crew Member can participate in multiple Launches.
- Crew Members have a nationality linking them to a Country.

1. Continent to Country:
 - Relationship: One-to-Many
2. Country to Launchpad:
 - Relationship: One-to-Many
3. Launchpad to Launch:
 - Relationship: One-to-Many
4. Launch to LaunchVehicle:
 - Relationship: Many-to-One
5. LaunchVehicle to LaunchVehicleType:
 - Relationship: Many-to-One
6. LaunchVehicleType to Manufacturer:
 - Relationship: Many-to-One
7. Launch to Payload:
 - Relationship: One-to-One
8. Payload to Manufacturer:
 - Relationship: Many-to-One
9. Launch to LaunchCrew:
 - Relationship: One-to-Many
10. CrewMember to LaunchCrew:
 - Relationship: One-to-Many
11. CrewMember to Country:
 - Relationship: Many-to-One

Relation Schemas

1. Launchpad(LaunchpadID, Name, Location, CountryID)
2. Country (CountryID, Name, ContinentID)
3. Continent (ContinentID, Name)
4. Launch (LaunchID, Date, Time, Name, Crewed(Boolean), LaunchpadID, VehicleID, PayloadID)
5. Launch Vehicle (VehicleID, SerialNumber, VehicleTypeID)

6. Launch Vehicle Type (VehicleTypeID, Name, MaxThrust, Reusable(Boolean), ManufacturerID)
7. Manufacturer(ManufacturerID,Name)
8. Payload(PayloadID, Description, ManufacturerID)
9. Crew Member (CrewMemberID , Name, Surname, CountryID)
10. Launch Crew (LaunchID, CrewMemberID, CompositePK (LaunchID, CrewMemberID))

RD Diagram

